Signals and Systems — Reference

notation: CT and DT side by side

Item	Continuous time (CT)	Discrete time (DT)
signal	$x(t)$, round brackets	$x[n]$, square brackets
time variable	t, any real number	n, integers only
period	T (seconds)	N (samples, an integer)
frequency	ω (rad/s), f (Hz)	$\hat{\omega}$ (rad/sample)
energy, power	integral $\int$	sum $\sum$
rate of change	derivative $\frac{d}{dt}$	first difference $x[n]-x[n-1]$
accumulate	integral $\int_{-\infty}^{t}$	running sum $\sum_{k=-\infty}^{n}$
impulse	$\delta(t)$, area 1	$\delta[n]$, height 1

sinusoid quantities

Symbol	Name	Unit	Link
A	amplitude	same as x	peak value
ω	angular frequency	rad/s	$\omega = 2\pi f$
f	frequency	Hz = cycles/s	$f = \frac{1}{T}$
T	period	s	$T = \frac{2\pi}{\omega}$
ϕ	phase	rad	$\phi = -\omega t_0$
t_0	time shift	s	$t_0 = -\frac{\phi}{\omega}$
$\hat{\omega}$	DT frequency	rad/sample	$\hat{\omega} = \omega T_s$
T_s, f_s	sampling period, rate	s, Hz	$f_s = \frac{1}{T_s}$

trig and complex number facts

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0

sine as a cosine	$\sin\theta = \cos\left(\theta - \frac{\pi}{2}\right)$
negative amplitude	$-A\cos\theta = A\cos(\theta+\pi)$
square of a cosine	$\cos^2\theta = \frac{1+\cos(2\theta)}{2}$
Euler's formula	$e^{j\theta} = \cos\theta + j\sin\theta$
cosine from exponentials	$\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$
sine from exponentials	$\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$
size of $a+jb$	$\sqrt{a^2+b^2}$
angle of $a+jb$	$\arctan\frac{b}{a}$ (check the quadrant)
polar form	$C = \lvert C\rvert e^{j\phi}$
size of $e^{j\theta}$	1
whole turns	$e^{j2\pi k} = 1$ for each integer k

sums and integrals

Item	Result
infinite geometric sum, $\lvert r\rvert < 1$	$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$
finite geometric sum, $r \neq 1$	$\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r}$
count of terms from $-N$ to N	$2N+1$
$\int t^n\,dt$	$\frac{t^{n+1}}{n+1}$
$\int e^{at}\,dt$	$\frac{e^{at}}{a}$
$\int_0^{\infty} e^{-at}\,dt, \quad a>0$	$\frac{1}{a}$
definite integral	$F(\text{top}) - F(\text{bottom})$

formula sheet

1. Signals and sampling

sampling	$x[n] = x(nT_s)$
sampling rate	$f_s = \frac{1}{T_s}$

2–3. Sinusoids

CT sinusoid	$x(t) = A\cos(\omega t + \phi)$
angular frequency	$\omega = 2\pi f$
period	$T = \frac{1}{f} = \frac{2\pi}{\omega}$
phase → time shift	$t_0 = -\frac{\phi}{\omega}$
time shift → phase	$\phi = -\omega t_0$
DT sinusoid	$x[n] = A\cos(\hat{\omega} n + \phi)$
CT → DT frequency	$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s}$
DT → CT frequency	$f = \frac{\hat{\omega} f_s}{2\pi}$
aliases	$\hat{\omega} + 2\pi k, \quad f + k f_s$

4–5. Energy and power

CT energy	$E_\infty = \int_{-\infty}^{\infty} \lvert x(t)\rvert^2\,dt$
DT energy	$E_\infty = \sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2$
CT power	$P_\infty = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \lvert x(t)\rvert^2\,dt$
DT power	$P_\infty = \lim_{N\to\infty} \frac{1}{2N+1}\sum_{n=-N}^{N} \lvert x[n]\rvert^2$
periodic CT power	$P = \frac{1}{T}\int_{\text{one period}} \lvert x(t)\rvert^2\,dt$
periodic DT power	$P = \frac{1}{N}\sum_{\text{one period}} \lvert x[n]\rvert^2$
sinusoid power	$P = \frac{A^2}{2}$
constant c power	$P = c^2$
flat piece, level A, length D	$A^2 D$
ramp $0 \to A$, length D	$\frac{A^2 D}{3}$
line $A \to B$, length D	$\frac{D}{3}\left(A^2 + AB + B^2\right)$
DT stem of height h	h^2

6. Transformations

shift	$x(t-t_0)$: right if $t_0 > 0$
reversal	$x(-t)$: mirror about $t = 0$
scaling	$x(at)$: compress if $\lvert a\rvert > 1$
point map for $x(at+b)$	$t_{\text{new}} = \frac{t_{\text{old}} - b}{a}$
undo $y(t) = x(at+b)$	$x(t) = y\left(\frac{t-b}{a}\right)$

7. Periodic signals

CT test	$x(t+T) = x(t)$
DT test	$x[n+N] = x[n]$, N an integer
DT sinusoid period	$\frac{\hat{\omega}}{2\pi} = \frac{m}{N}$ in lowest terms
sum of two periodic signals	lcm of the two periods

8. Even and odd

even test	$x(-t) = x(t)$
odd test	$x(-t) = -x(t)$
even part	$\mathcal{E}v\{x\} = \frac{x(t)+x(-t)}{2}$
odd part	$\mathcal{O}d\{x\} = \frac{x(t)-x(-t)}{2}$

9. Complex exponentials

CT form	$x(t) = Ce^{at}, \quad a = \sigma + j\omega$
real part	$\lvert C\rvert e^{\sigma t}\cos(\omega t + \phi), \quad \phi = \angle C$
DT form	$x[n] = C\alpha^n$

10. Impulse and step

DT step → impulse	$\delta[n] = u[n] - u[n-1]$
DT impulse → step	$u[n] = \sum_{k=-\infty}^{n} \delta[k]$
CT step → impulse	$\delta(t) = \dfrac{du(t)}{dt}$
CT impulse → step	$u(t) = \int_{-\infty}^{t} \delta(\tau)\, d\tau$
sifting	$\int_{-\infty}^{\infty} x(t)\, \delta(t - t_0)\, dt = x(t_0)$
DT sifting	$\sum_n x[n]\, \delta[n - n_0] = x[n_0]$
pulse from steps	$u(t - t_1) - u(t - t_2)$
any DT signal as impulses	$x[n] = \sum_k x[k]\, \delta[n - k]$
ramp	$r(t) = t\, u(t), \quad r[n] = n\, u[n]$
ramp and step, CT	$\dfrac{dr}{dt} = u(t)$
ramp and step, DT	$r[n] - r[n-1] = u[n-1]$
graph from ramps	corner: $+(\Delta\text{slope})\, r(t - t_0)$

11–12. DT systems

unit delay D	$x[n] \to x[n-1]$
first difference	$y[n] = x[n] - x[n-1]$
3-point moving average	$y[n] = \frac{1}{3}\big(x[n] + x[n-1] + x[n-2]\big)$
accumulator	$y[n] = y[n-1] + x[n]$
bounded input	$\lvert x[n] \rvert \le B < \infty$
stable if	$\lvert y[n] \rvert \le$ a finite number

system property tests

Property	Test	Yes / no example
memoryless	y at n uses only x at n	yes x^2 no $x[n-1]$
causal	y at n uses no x at a later time	yes $x[n-1]$ no $x[n+1]$
stable	bounded x gives bounded y	yes x^2 no $\sum_{k \le n} x[k]$
invertible	you can solve for x from y	yes $2x$ no x^2
time invariant	delay in, then system = system, then delay out	yes $x[n-1]$ no $n\,x[n]$
linear	$ax_1 + bx_2 \to ay_1 + by_2$	yes $n\,x[n]$ no $2x + 3$

which tool? — given → find

Given → find	Do this	
$x(t)$, $T_s \to x[n]$	put $t = nT_s$	(1)
$A\cos(\omega t + \phi) \to f$, T, t_0	read, then divide	(2)
f, $f_s \to \hat{\omega}$	$\hat{\omega} = 2\pi f / f_s$	(3)
dies out or finite length	energy E_∞	(4)
periodic or constant	power P_∞	(4)
a graph → energy	square each piece, add	(5)
$x(t) \to x(at + b)$	map each breakpoint	(6)
$y = x(at + b) \to x$	$x(t) = y((t - b)/a)$	(6)
two graphs → $y = A\,x(at + b)$	match two points, solve	(6)
signal → period	$2\pi/\omega$, or reduce $\hat{\omega}/2\pi$	(7)
signal → even, odd, neither	compare $x(-t)$ with $x(t)$	(8)
Ce^{at} → real signal	Euler, then the real part	(9)
integral with $\delta(t - t_0)$	sift: evaluate at t_0	(10)
a pulse graph → formula	a step up, a step down	(10)
DT stems → impulses	one $x[k]\,\delta[n-k]$ per stem	(10)
graph with corners → formula	a ramp per corner	(10)
blocks → one equation	substitute block by block	(11)
equation ↔ block diagram	one D per shift	(11)
system → properties	test each one	(12)
system → stable?	bound with B	(12)

The number in parentheses is the concept section.

Signals and Systems — Concepts

1. signals and sampling

A **signal** is a function that carries information. The independent variable is usually time.

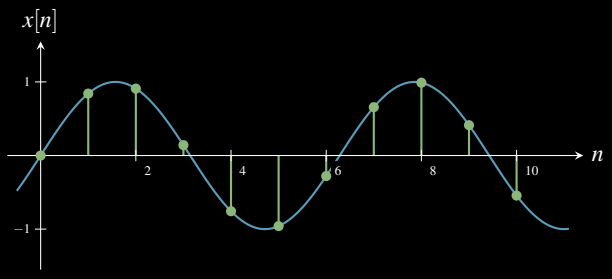
- **Continuous time** (CT): $x(t)$. t can be any real number.
- **Discrete time** (DT): $x[n]$. n is an integer only.

$x[2.5]$ does not exist. Between two samples, a DT signal is not zero. It is undefined.

Sampling makes a DT signal from a CT signal. Take one value each T_s seconds:

$$t = nT_s \quad \Rightarrow \quad x[n] = x(nT_s)$$

T_s is the **sampling period**. $f_s = \frac{1}{T_s}$ is the **sampling rate** (samples per second).



The curve is the CT signal. The stems are the samples.

Worked: sample $x(t) = t^2$ with $T_s = 0.5$ s

1. PUT $t = nT_s$
$x[n] = x(0.5n) = (0.5n)^2$

↓

2. SIMPLIFY
$(0.5n)^2 = 0.5^2 \cdot n^2 = 0.25n^2$

↓

3. EVALUATE EACH SAMPLE
$x[0] = 0, \quad x[1] = 0.25, \quad x[2] = 0.25 \cdot 4 = 1, \quad x[3] = 0.25 \cdot 9 = 2.25$

↓

4. CHECK ONE SAMPLE
$n = 2$ is $t = 2 \cdot 0.5 = 1$ s. $\quad x(1) = 1^2 = 1$ ✓

Worked: (reverse) $x[n] = 0.25n^2$ came from $T_s = 0.5$ s. Find $x(t)$

1. SOLVE $t = nT_s$ FOR n
$$n = \frac{t}{T_s} = \frac{t}{0.5} = 2t$$

↓

2. SUBSTITUTE
$x(t) = 0.25(2t)^2$

↓

3. SIMPLIFY
$0.25 \cdot 4t^2 = t^2$

2. sinusoids

$$x(t) = A\cos(\omega t + \phi)$$

- A: **amplitude**. The peak value. The signal swings between $-A$ and A.
- ω: **angular frequency** in rad/s. $\omega = 2\pi f$.
- f: **frequency** in Hz (cycles per second).
- $T = \frac{1}{f} = \frac{2\pi}{\omega}$: the **period** in seconds.
- ϕ: **phase** in radians. It slides the wave along the time axis.

Phase as a time shift:

$$A\cos(\omega t + \phi) = A\cos\big(\omega(t - t_0)\big), \qquad t_0 = -\frac{\phi}{\omega}$$

A negative ϕ gives $t_0 > 0$: the wave is **delayed** (it moves right).

Put the signal in the form $A\cos(\omega t + \phi)$ before you read the parts. Change a sine with $\sin\theta = \cos(\theta - \frac{\pi}{2})$. Change a negative A with $-A\cos\theta = A\cos(\theta + \pi)$.

Which way? — between the sinusoid numbers

Given → find	Do this
$\omega \to f$	divide by 2π: $f = \frac{\omega}{2\pi}$
$f \to \omega$	multiply by 2π: $\omega = 2\pi f$
$f \to T$	reciprocal: $T = \frac{1}{f}$
$T \to \omega$	$\omega = \frac{2\pi}{T}$
$\phi \to t_0$	$t_0 = -\frac{\phi}{\omega}$
$t_0 \to \phi$	$\phi = -\omega t_0$

Worked: read the parts of $x(t) = 3\cos\left(4\pi t - \frac{\pi}{3}\right)$

1. MATCH THE FORM $A\cos(\omega t + \phi)$
$A = 3, \quad \omega = 4\pi$ rad/s, $\quad \phi = -\frac{\pi}{3}$ rad

↓

2. FREQUENCY
$$f = \frac{\omega}{2\pi} = \frac{4\pi}{2\pi} = 2 \text{ Hz}$$

↓

3. PERIOD
$$T = \frac{1}{f} = \frac{1}{2} = 0.5 \text{ s}$$

↓

4. TIME SHIFT
$$t_0 = -\frac{\phi}{\omega} = -\frac{-\pi/3}{4\pi} = \frac{\pi/3}{4\pi} = \frac{1}{3 \cdot 4} = \tfrac{1}{12} \text{ s}$$

↓

5. REWRITE AS A SHIFT
$x(t) = 3\cos\left(4\pi(t - \frac{1}{12})\right)$: delayed by $\frac{1}{12}$ s

↓

6. CHECK
$4\pi \cdot \frac{1}{12} = \frac{\pi}{3}$. So $4\pi(t - \frac{1}{12}) = 4\pi t - \frac{\pi}{3}$ ✓

Worked: (sine first) read the parts of $x(t) = 4\sin(10t)$

1. CHANGE SINE TO COSINE
$\sin\theta = \cos(\theta - \frac{\pi}{2})$, so $x(t) = 4\cos\left(10t - \frac{\pi}{2}\right)$

↓

2. MATCH THE FORM
$A = 4, \quad \omega = 10$ rad/s, $\quad \phi = -\frac{\pi}{2}$

↓

3. FREQUENCY AND PERIOD
$$f = \frac{10}{2\pi} = 1.59 \text{ Hz}. \qquad T = \frac{2\pi}{10} = 0.628 \text{ s}$$

Worked: (reverse) build $x(t)$ from $A = 2$, $T = 0.4$ s, first peak at $t = 0.1$ s

1. PERIOD TO ω
$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.4} = 5\pi \text{ rad/s}$$

↓

2. PEAK TIME IS THE SHIFT
A cosine peaks where its angle is 0. The first peak is at $t = 0.1$, so $t_0 = 0.1$ s.

↓

3. SHIFT TO PHASE
$\phi = -\omega t_0 = -5\pi \cdot 0.1 = -0.5\pi = -\frac{\pi}{2}$

↓

4. ASSEMBLE
$x(t) = 2\cos\left(5\pi t - \frac{\pi}{2}\right)$

↓

5. CHECK AT $t = 0.1$
Angle: $5\pi(0.1) - \frac{\pi}{2} = \frac{\pi}{2} - \frac{\pi}{2} = 0$. $\quad 2\cos 0 = 2$, the peak ✓

3. DT sinusoids and DT frequency

Sample $A\cos(\omega t+\phi)$ at $t=nT_s$:

$$x[n]=A\cos(\omega nT_s+\phi)=A\cos(\hat{\omega}n+\phi)$$

$\hat{\omega}=\omega T_s=\dfrac{2\pi f}{f_s}$ is the **DT frequency**, in radians per sample.

Which way? — CT frequency or DT frequency

Given → find	Do this
$f, f_s \to \hat{\omega}$	$\hat{\omega}=\dfrac{2\pi f}{f_s}$
$\omega, T_s \to \hat{\omega}$	$\hat{\omega}=\omega T_s$
$\hat{\omega}, f_s \to f$	$f=\dfrac{\hat{\omega} f_s}{2\pi}$ (the lowest answer)

DT to CT has more than one answer. $\hat{\omega}$ and $\hat{\omega}+2\pi$ give the same samples. So f and $f+f_s$ give the same samples. This is **aliasing**.

Worked: sample $x(t)=\cos(200\pi t)$ at $f_s=800$ Hz

1. READ ω AND f
$\omega=200\pi$ rad/s. $\quad f=\dfrac{200\pi}{2\pi}=100$ Hz

↓

2. SAMPLING PERIOD
$T_s=\dfrac{1}{800}$ s

↓

3. DT FREQUENCY
$\hat{\omega}=\omega T_s=\dfrac{200\pi}{800}=\frac{\pi}{4}$ rad/sample

↓

4. THE DT SIGNAL
$x[n]=\cos\left(\frac{\pi}{4}n\right)$

Worked: (reverse) $\hat{\omega}=\frac{\pi}{4}$ at $f_s=800$ Hz. Find f

1. FORMULA
$f=\dfrac{\hat{\omega} f_s}{2\pi}$

↓

2. SUBSTITUTE
$f=\dfrac{(\pi/4)\cdot 800}{2\pi}=\dfrac{200\pi}{2\pi}=100$ Hz

Worked: the alias: does $f=900$ Hz give the same samples?

1. DT FREQUENCY
$\hat{\omega}=\dfrac{2\pi\cdot 900}{800}=\dfrac{1800\pi}{800}=\dfrac{9\pi}{4}$

↓

2. REMOVE A WHOLE TURN
$\dfrac{9\pi}{4}-2\pi=\dfrac{9\pi}{4}-\dfrac{8\pi}{4}=\dfrac{\pi}{4}$

↓

3. COMPARE
Same $\hat{\omega}$ as 100 Hz ⇒ same samples. $\quad 900=100+800=f+f_s$.

4. energy and power

Treat x as a voltage across a $1\,\Omega$ resistor. Then $|x|^2$ is the power at each instant.

Total energy:

$$E_\infty=\int_{-\infty}^{\infty}|x(t)|^2\,dt \qquad E_\infty=\sum_{n=-\infty}^{\infty}|x[n]|^2$$

Average power:

$$P_\infty=\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}|x(t)|^2\,dt \qquad P_\infty=\lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N}|x[n]|^2$$

Periodic shortcut: average over one period: $P=\frac{1}{T}\int_{\text{one period}}|x(t)|^2\,dt$.

Which one to compute?

The signal	Result
dies out, or has finite length	**energy signal**: E_∞ finite, $P_\infty=0$
periodic, or constant	**power signal**: P_∞ finite, $E_\infty=\infty$
grows without limit ($x=t$)	**neither**: $E_\infty=\infty$, $P_\infty=\infty$

Square first. The energy is the area under $|x|^2$, not under x.

Worked: pulse: $x(t)=2$ for $0\le t\le 3$, 0 elsewhere

1. SQUARE
$|x(t)|^2=2^2=4$ on $[0,3]$. 0 elsewhere.

↓

2. LIMITS SHRINK TO WHERE $x\neq 0$
$E_\infty=\int_0^3 4\,dt$

↓

3. ANTIDERIVATIVE
$F(t)=4t$

↓

4. TOP MINUS BOTTOM
$4(3)-4(0)=12$

↓

5. POWER
$P_\infty=\lim_{T\to\infty}\dfrac{12}{2T}=0\Rightarrow$ energy signal

Worked: decaying exponential: $x(t)=3e^{-2t}u(t)$

1. WHERE IS IT NOT ZERO?
$u(t)=0$ for $t<0$. So integrate from 0 to ∞.

↓

2. SQUARE
$\left(3e^{-2t}\right)^2=3^2\cdot\left(e^{-2t}\right)^2=9e^{-4t}$ *(double the exponent)*

↓

3. SET UP
$E_\infty=\lim_{b\to\infty}\int_0^b 9e^{-4t}\,dt$

↓

4. ANTIDERIVATIVE
$F(t)=\dfrac{9}{-4}e^{-4t}=-2.25e^{-4t}$

↓

5. TOP ($t=b$), THEN $b\to\infty$
$F(b)=-2.25e^{-4b}\to -2.25(0)=0$

↓

6. BOTTOM ($t=0$)
$F(0)=-2.25e^0=-2.25(1)=-2.25$

↓

7. TOP MINUS BOTTOM
$E_\infty=0-(-2.25)=2.25\quad\Rightarrow$ energy signal

Worked: DT geometric: $x[n]=\left(\frac{1}{2}\right)^n u[n]$

1. WHERE IS IT NOT ZERO?
$u[n]=0$ for $n<0$. So sum from $n=0$.

↓

2. SQUARE
$\left(\left(\frac{1}{2}\right)^n\right)^2=\left(\frac{1}{2}\right)^{2n}=\left(\frac{1}{4}\right)^n$

↓

3. SET UP
$E_\infty=\sum_{n=0}^{\infty}\left(\frac{1}{4}\right)^n$ *(a geometric sum with $r=\frac{1}{4}$)*

↓

4. GEOMETRIC SUM FORMULA
$\dfrac{1}{1-r}=\dfrac{1}{1-\frac{1}{4}}=\dfrac{1}{\frac{3}{4}}=\frac{4}{3}$

↓

5. CHECK WITH THE FIRST TERMS
$1+0.25+0.0625+0.0156=1.328$, close to 1.333 ✓

Worked: sinusoid: $x(t) = 4\cos(3t)$

1. SQUARE
$|x|^2 = 16\cos^2(3t)$

↓

2. USE $\cos^2\theta = \frac{1+\cos 2\theta}{2}$
$16 \cdot \dfrac{1+\cos(6t)}{2} = 8 + 8\cos(6t)$

↓

3. AVERAGE EACH TERM
The constant 8 averages to 8. $8\cos(6t)$ averages to 0 *(equal area above and below 0)*.

↓

4. POWER
$P_\infty = 8 + 0 = 8$

↓

5. CHECK WITH THE SHORTCUT
$\dfrac{A^2}{2} = \dfrac{4^2}{2} = \dfrac{16}{2} = 8$ ✓ $E_\infty = \infty \Rightarrow$ power signal

Worked: constant: $x(t) = 3$ ***for all*** t

1. SQUARE
$|x|^2 = 9$

↓

2. INTEGRAL OVER $[-T, T]$
$\int_{-T}^{T} 9\,dt = 9T - 9(-T) = 18T$

↓

3. DIVIDE BY $2T$
$\dfrac{18T}{2T} = 9$

↓

4. LET $T \to \infty$
$P_\infty = 9$ *(no T is left).* $E_\infty = lim\, 18T = \infty$

Worked: DT power: $x[n] = (-1)^n$

1. SQUARE
$\left((-1)^n\right)^2 = 1$ for each n

↓

2. SUM FROM $-N$ **TO** N
$2N+1$ terms, each 1: sum $= 2N+1$

↓

3. DIVIDE AND LET $N \to \infty$
$P_\infty = \dfrac{2N+1}{2N+1} = 1$

5. energy from a graph

Most graph problems use flat pieces and straight lines. Do not set up one big integral. Work piece by piece.

1. **Read the graph into pieces.** Write $x(t)$ on each time range: a constant, or a straight line.
2. **Square each piece.**
3. **CT**: integrate each squared piece over its range. **DT**: square each stem height.
4. **Add the pieces.**

Equation of a straight line through (t_1, x_1) with slope $m = \frac{\Delta x}{\Delta t}$:
$x(t) = x_1 + m(t - t_1)$.

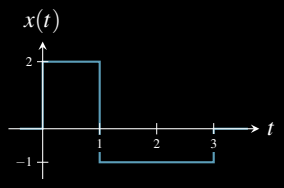
Shortcuts for one piece of length D:

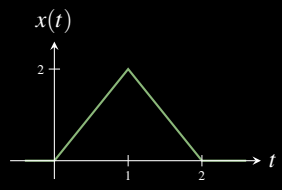
Piece of the graph	Energy of that piece
flat at level A	$A^2 D$
ramp from 0 to A (or A to 0)	$\dfrac{A^2 D}{3}$
line from A to B	$\dfrac{D}{3}(A^2 + AB + B^2)$
DT stem of height h	h^2

Negative parts count positive. $(-2)^2 = 4$. They do not cancel positive parts.

A squared ramp is a curve, not a triangle. Use $\frac{A^2 D}{3}$, not $\frac{A^2 D}{2}$.

If the graph repeats forever, $E_\infty = \infty$. Find the power over one period.





Worked: two flat pieces (left graph)

1. READ THE PIECES
$x(t) = 2$ on $[0,1]$. $x(t) = -1$ on $[1,3]$. 0 elsewhere.

↓

2. SQUARE EACH PIECE
$2^2 = 4$ on $[0,1]$. $(-1)^2 = 1$ on $[1,3]$.

↓

3. ENERGY OF PIECE 1
$\int_0^1 4\,dt = 4 \cdot (1-0) = 4$

↓

4. ENERGY OF PIECE 2
$\int_1^3 1\,dt = 1 \cdot (3-1) = 2$

↓

5. ADD
$E_\infty = 4 + 2 = 6$

Worked: a triangle (right graph)

1. RISE: FIND THE LINE
From $(0,0)$ to $(1,2)$: slope $m = \frac{2-0}{1-0} = 2$. $x(t) = 0 + 2(t-0) = 2t$ on $[0,1]$.

↓

2. RISE: SQUARE
$(2t)^2 = 4t^2$

↓

3. RISE: INTEGRATE
$F(t) = \dfrac{4t^3}{3}$. $F(1) - F(0) = \dfrac{4}{3} - 0 = \dfrac{4}{3}$

↓

4. FALL: FIND THE LINE
From $(1,2)$ to $(2,0)$: slope $m = \frac{0-2}{2-1} = -2$. $x(t) = 2 - 2(t-1) = 4 - 2t$ on $[1,2]$.

↓

5. FALL: SQUARE
$(4-2t)^2 = 16 - 16t + 4t^2$ *($(a-b)^2 = a^2 - 2ab + b^2$)*

↓

6. FALL: ANTIDERIVATIVE
$G(t) = 16t - 8t^2 + \dfrac{4t^3}{3}$

↓

7. FALL: TOP *($t = 2$)*
$G(2) = 32 - 32 + \dfrac{32}{3} = \dfrac{32}{3}$

↓

8. FALL: BOTTOM *($t = 1$)*
$G(1) = 16 - 8 + \dfrac{4}{3} = 8 + \dfrac{4}{3} = \dfrac{28}{3}$

↓

9. FALL: TOP MINUS BOTTOM
$\dfrac{32}{3} - \dfrac{28}{3} = \dfrac{4}{3}$

↓

10. ADD
$E_\infty = \dfrac{4}{3} + \dfrac{4}{3} = \frac{8}{3}$

↓

11. CHECK WITH THE SHORTCUT
Each ramp: $\dfrac{A^2 D}{3} = \dfrac{2^2 \cdot 1}{3} = \dfrac{4}{3}$ ✓

Worked: a line that does not start at 0: from 1 to 3 over $[0,2]$

1. FIND THE LINE
Slope $m = \frac{3-1}{2-0} = 1$. $x(t) = 1 + 1 \cdot (t-0) = 1+t$

↓

2. SQUARE
$(1+t)^2 = 1 + 2t + t^2$

↓

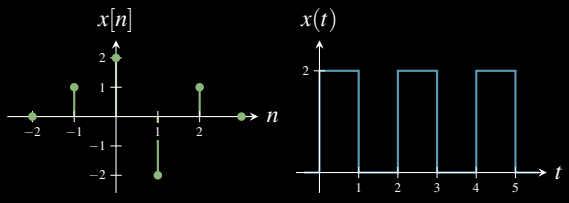
3. ANTIDERIVATIVE
$F(t) = t + t^2 + \dfrac{t^3}{3}$

↓

4. TOP MINUS BOTTOM
$F(2) = 2 + 4 + \dfrac{8}{3} = \dfrac{26}{3}$. $F(0) = 0$. $E = \frac{26}{3}$

↓

5. CHECK WITH THE SHORTCUT
$\dfrac{D}{3}\left(A^2 + AB + B^2\right) = \dfrac{2}{3}(1+3+9) = \dfrac{26}{3}$ ✓



Worked: a DT stem plot (left graph)

1. READ THE STEMS
$x[-1] = 1$, $x[0] = 2$, $x[1] = -2$, $x[2] = 1$. All others 0.

↓

2. SQUARE EACH STEM
$1^2 = 1$, $2^2 = 4$, $(-2)^2 = 4$, $1^2 = 1$

↓

3. ADD
$E_\infty = 1 + 4 + 4 + 1 = 10$ *(no integral: DT energy is a sum)*

Worked: a periodic graph (right graph): power, not energy

1. ONE PERIOD
The pattern repeats every $T = 2$ s: $x = 2$ on $[0,1]$, $x = 0$ on $[1,2]$.

↓

2. ENERGY OF ONE PERIOD
$2^2 \cdot 1 + 0^2 \cdot 1 = 4$

↓

3. DIVIDE BY THE PERIOD
$P = \dfrac{4}{2} = 2$

↓

4. TOTAL ENERGY
It repeats forever: $E_\infty = \infty \Rightarrow$ power signal

6. transformations of the time variable

- **Time shift**: $x(t - t_0)$. If $t_0 > 0$, the signal moves right (delay). If $t_0 < 0$, it moves left (advance).
- **Time reversal**: $x(-t)$. Mirror the signal about $t = 0$.
- **Time scaling**: $x(at)$. If $|a| > 1$, the signal is compressed. If $|a| < 1$, it is stretched.

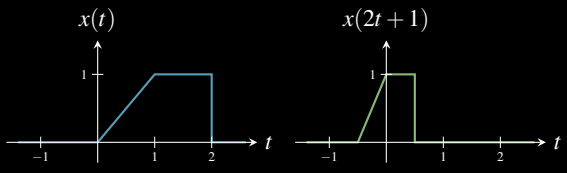
Point-map method for $x(at + b)$: each breakpoint of x moves to a new time, and keeps its value.

$$t_{\text{new}} = \frac{t_{\text{old}} - b}{a}$$

This comes from solving $at_{\text{new}} + b = t_{\text{old}}$. It gives the answer in one step, with no order of operations to remember.

Two-step method: first shift by b to make $x(t + b)$. Then replace t with at. Scale first, then shift by b, is wrong. $x(2t)$ shifted left by 1 is $x(2(t+1)) = x(2t+2)$, not $x(2t+1)$.

DT scaling loses samples: $x[2n]$ keeps only the even-index samples.



Worked: sketch $x(2t+1)$ from $x(t)$ (left graph)

1. LIST THE BREAKPOINTS OF x
$t = 0$ (value 0), $t = 1$ (value 1), $t = 2$ (value 1, then drops to 0)

↓

2. READ a AND b
$x(2t+1)$: $a = 2$, $b = 1$

↓

3. MAP $t = 0$
$\dfrac{0-1}{2} = -\dfrac{1}{2}$

↓

4. MAP $t = 1$
$\dfrac{1-1}{2} = 0$

↓

5. MAP $t = 2$
$\dfrac{2-1}{2} = \dfrac{1}{2}$

↓

6. DRAW WITH THE SAME VALUES
Ramp $0 \to 1$ on $[-\frac{1}{2}, 0]$. Flat at 1 on $[0, \frac{1}{2}]$. 0 elsewhere (right graph).

↓

7. CHECK ONE POINT
At $t = 0$: $x(2 \cdot 0 + 1) = x(1) = 1$ ✓

Worked: (reversal) sketch $x(-t+2)$ from the same $x(t)$

1. READ a AND b
$a = -1$, $b = 2$. $t_{\text{new}} = \dfrac{t_{\text{old}} - 2}{-1} = 2 - t_{\text{old}}$

↓

2. MAP EACH BREAKPOINT
$0 \to 2$, $1 \to 1$, $2 \to 0$

↓

3. DRAW WITH THE SAME VALUES
Flat at 1 on $[0,1]$. Ramp $1 \to 0$ on $[1,2]$. The picture is mirrored.

↓

4. CHECK ONE POINT
At $t = 1.5$: $x(-1.5+2) = x(0.5) = 0.5$. The ramp from 1 to 0 is at 0.5 there ✓

Worked: DT: $x[0] = 1$, $x[1] = 2$, $x[2] = 3$, else 0

1. DELAY $x[n-2]$
Each sample moves right 2: $n = 2$: 1, $n = 3$: 2, $n = 4$: 3

↓

2. REVERSE $x[-n]$
Each sample mirrors: $n = 0$: 1, $n = -1$: 2, $n = -2$: 3

↓

3. COMPRESS $x[2n]$
$n = 0$: $x[0] = 1$. $n = 1$: $x[2] = 3$. $x[1] = 2$ is lost.

Which way? — undo a transformation

Given → find	Do this
$x(t) \to x(at+b)$	map each breakpoint: $t_{\text{new}} = \frac{t_{\text{old}} - b}{a}$
$y(t) = x(at+b) \to x(t)$	$x(t) = y\left(\frac{t-b}{a}\right)$

$x[2n]$ cannot be undone. The odd-index samples are lost.

Worked: (reverse) $y(t) = x(2t+1)$ is given. Find $x(t)$

1. NAME THE INSIDE
Let $\tau = 2t+1$. Then $y(t) = x(\tau)$.

↓

2. SOLVE FOR t
$2t = \tau - 1 \Rightarrow t = \dfrac{\tau - 1}{2}$

↓

3. SUBSTITUTE
$x(\tau) = y\left(\dfrac{\tau-1}{2}\right)$. Rename τ as t: $x(t) = y\left(\frac{t-1}{2}\right)$

↓

4. CHECK
$y\left(\frac{t-1}{2}\right) = x\left(2\cdot\frac{t-1}{2}+1\right) = x(t-1+1) = x(t)$ ✓

Which way? — find the transformation from x to y

Some problems give two graphs and ask for y in terms of x:
$y(t) = A\,x(at+b)$.

1. **Amplitude**: compare the heights. $A = \frac{\text{height of } y}{\text{height of } x}$. A flipped shape (upside down) gives a negative A.
2. **Orientation**: is the shape the same way round, or mirrored? Mirrored means $a < 0$.
3. **Two matching points**: pick two breakpoints of x (t_1, t_2) and the points of y they move to (τ_1, τ_2).
4. **Solve** $a\tau_1 + b = t_1$ and $a\tau_2 + b = t_2$ for a and b. *(y at τ is x at $a\tau+b$)*
5. **Check** a third point.

Worked: $x(t)$ is a ramp from 0 to 1 on $[0,4]$. $y(t)$ is the same ramp, height 3, on $[-1,1]$

1. AMPLITUDE
$A = \dfrac{3}{1} = 3$

↓

2. ORIENTATION
Both rise from left to right ⇒ not mirrored, $a > 0$

↓

3. MATCHING POINTS
The start $t_1 = 0$ moves to $\tau_1 = -1$. The end $t_2 = 4$ moves to $\tau_2 = 1$.

↓

4. TWO EQUATIONS
$a(-1) + b = 0$ and $a(1) + b = 4$

↓

5. SUBTRACT THE FIRST FROM THE SECOND
$a - (-a) = 4 - 0 \Rightarrow 2a = 4 \Rightarrow a = 2$

↓

6. BACK-SUBSTITUTE
$-2 + b = 0 \Rightarrow b = 2$

↓

7. RESULT AND CHECK
$y(t) = 3x(2t+2)$. Middle: $t = 0$ gives $x(2) = \frac{1}{2}$, so $y = 1.5$, half of 3 ✓

Worked: (mirrored) the same x, but y falls from 3 to 0 on $[-1,1]$

1. MATCHING POINTS
The start of the ramp ($t_1 = 0$, value 0) is now on the right: $\tau_1 = 1$. The end ($t_2 = 4$) is on the left: $\tau_2 = -1$.

↓

2. TWO EQUATIONS
$a(1) + b = 0$ and $a(-1) + b = 4$

↓

3. SOLVE
Subtract: $2a = -4 \Rightarrow a = -2$. Then $b = 2$.

↓

4. RESULT
$y(t) = 3x(-2t+2)$ *($a < 0$ because the shape is mirrored)*

Worked: (DT) $x[0] = 1$, $x[1] = 2$, $x[2] = 3$. $y[-1] = 3$, $y[0] = 2$, $y[1] = 1$. Find y in terms of x

1. ORIENTATION
x rises to the right. y falls to the right ⇒ mirrored: $y[n] = x[-n+b]$

↓

2. ONE MATCHING POINT
$y[-1] = 3 = x[2]$. So $-(-1)+b = 2 \Rightarrow 1+b = 2 \Rightarrow b = 1$

↓

3. RESULT
$y[n] = x[-n+1]$

↓

4. CHECK THE OTHERS
$n = 0$: $x[1] = 2$ ✓ $n = 1$: $x[0] = 1$ ✓

7. periodic signals

A signal is **periodic** if a shift by the period does not change it:

$$x(t) = x(t+T) \qquad x[n] = x[n+N]$$

The smallest positive T (or N) is the **fundamental period**. N must be an integer.

- **CT sinusoid** $\cos(\omega t)$: always periodic. $T = \frac{2\pi}{\omega}$.
- **DT sinusoid** $\cos(\hat{\omega} n)$: periodic only if $\frac{\hat{\omega}}{2\pi}$ is a fraction $\frac{m}{N}$ (rational). Reduce it to lowest terms. The denominator is N.
- **Sum of two periodic signals**: the period is the least common multiple (lcm) of the two periods.

$\cos(n)$ is not periodic. $\frac{1}{2\pi}$ is not a fraction of integers. The CT signal $\cos(t)$ is periodic. Sampling can destroy periodicity.

Worked: period of $\cos\left(\frac{2\pi}{3}t\right) + \sin\left(\frac{\pi}{2}t\right)$

1. FIRST PERIOD
$T_1 = \dfrac{2\pi}{2\pi/3} = 2\pi\cdot\dfrac{3}{2\pi} = 3$ s

↓

2. SECOND PERIOD
$T_2 = \dfrac{2\pi}{\pi/2} = 2\pi\cdot\dfrac{2}{\pi} = 4$ s

↓

3. LEAST COMMON MULTIPLE
Multiples of 3: $3, 6, 9, 12$. Multiples of 4: $4, 8, 12$. $T = 12$ s

↓

4. CHECK
12 s holds $\frac{12}{3} = 4$ cycles of the first and $\frac{12}{4} = 3$ of the second ✓

Worked: DT period of $x[n] = \cos\left(\frac{3\pi}{8}n\right)$

1. DIVIDE BY 2π
$\dfrac{\hat{\omega}}{2\pi} = \dfrac{3\pi/8}{2\pi} = \dfrac{3\pi}{8}\cdot\dfrac{1}{2\pi} = \dfrac{3}{16}$

↓

2. LOWEST TERMS?
3 and 16 share no factor. $N = 16$

↓

3. CHECK
$\dfrac{3\pi}{8}\cdot 16 = 6\pi = 3\cdot 2\pi$, whole turns ✓

Worked: reduce first: $x[n] = \cos\left(\frac{4\pi}{6}n\right)$

1. DIVIDE BY 2π
$\dfrac{4\pi/6}{2\pi} = \dfrac{4}{12}$

↓

2. REDUCE
$\dfrac{4}{12} = \dfrac{1}{3} \Rightarrow N = 3$ not 12

Worked: is $x[n] = \cos(n)$ periodic?

1. DIVIDE BY 2π
$\frac{1}{2\pi}$

↓

2. IS IT A FRACTION OF INTEGERS?
No: π is irrational. $\Rightarrow$ not periodic

8. even and odd signals

- **Even**: $x(-t) = x(t)$. Mirror image about $t = 0$. Example: cos, t^2.
- **Odd**: $x(-t) = -x(t)$. Point symmetry about the origin. Example: sin, t^3.

Test: replace t with $-t$, simplify, and compare with $x(t)$.
An odd signal is always 0 at the origin: $x(0) = -x(0)$ forces $x(0) = 0$.
Every signal is an even part plus an odd part:

$$\mathcal{E}v\{x\} = \frac{x(t)+x(-t)}{2} \qquad \mathcal{O}d\{x\} = \frac{x(t)-x(-t)}{2}$$

Products: even · even = even, odd · odd = even, even · odd = odd.

Worked: is $x(t) = t^3 - 2t$ even, odd, or neither?

1. REPLACE t WITH $-t$
$x(-t) = (-t)^3 - 2(-t)$

↓

2. SIMPLIFY
$(-t)^3 = -t^3$, $-2(-t) = 2t$. $x(-t) = -t^3 + 2t$

↓

3. COMPARE
$-t^3 + 2t = -(t^3 - 2t) = -x(t) \Rightarrow$ odd

Worked: $x(t) = t^2 + t$: test, then split

1. REPLACE t WITH $-t$
$x(-t) = (-t)^2 + (-t) = t^2 - t$

↓

2. COMPARE
$t^2 - t \neq x(t)$ and $\neq -x(t) \Rightarrow$ neither

↓

3. EVEN PART
$\frac{(t^2+t)+(t^2-t)}{2} = \frac{2t^2}{2} = t^2$

↓

4. ODD PART
$\frac{(t^2+t)-(t^2-t)}{2} = \frac{2t}{2} = t$

↓

5. CHECK
$t^2 + t = x(t)$ ✓

Worked: DT split: $x[0] = 1$, $x[1] = 2$, $x[2] = 3$, else 0

1. TABLE OF $x[n]$ AND $x[-n]$, THEN THE TWO PARTS

n	-2	-1	0	1	2
$x[n]$	0	0	1	2	3
$x[-n]$	3	2	1	0	0
$\frac{x[n]+x[-n]}{2}$	1.5	1	1	1	1.5
$\frac{x[n]-x[-n]}{2}$	−1.5	−1	0	1	1.5

↓

2. HOW TO FILL $x[-n]$
Read x at the opposite index. $n = -2$: $x[2] = 3$. $n = 1$: $x[-1] = 0$.

↓

3. CHECK EACH COLUMN
even + odd $= x[n]$. $n = 2$: $1.5 + 1.5 = 3$ ✓ $n = -2$: $1.5 - 1.5 = 0$ ✓

9. complex exponentials

Euler's formula: $e^{j\theta} = \cos\theta + j\sin\theta$. Engineers write $j = \sqrt{-1}$.
Back to cosine and sine: $\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$, $\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$.

CT form: $x(t) = Ce^{at}$, with $C = |C|e^{j\phi}$ and $a = \sigma + j\omega$.

Case	Real part looks like
a real ($\omega = 0$)	**real exponential**: grows if $\sigma > 0$, decays if $\sigma < 0$
$a = j\omega$ ($\sigma = 0$)	**sinusoid** $\lvert C\rvert\cos(\omega t + \phi)$, $T = \frac{2\pi}{\lvert\omega\rvert}$
general a	**damped sinusoid** $\lvert C\rvert e^{\sigma t}\cos(\omega t + \phi)$

The $|C|e^{\sigma t}$ part is the **envelope**.
DT form: $x[n] = C\alpha^n$. $|\alpha| < 1$: decays. $|\alpha| > 1$: grows. $\alpha < 0$: the sign alternates.
DT frequencies repeat every 2π: $e^{j(\hat{\omega}+2\pi)n} = e^{j\hat{\omega}n}e^{j2\pi n} = e^{j\hat{\omega}n}$, because n is an integer.

CT $e^{j\omega t}$	DT $e^{j\hat{\omega}n}$
each ω gives a different signal	$\hat{\omega}$ and $\hat{\omega}+2\pi$ give the same signal
always periodic	periodic only if $\frac{\hat{\omega}}{2\pi}$ is rational
larger $\omega \Rightarrow$ faster	fastest at $\hat{\omega} = \pi$ $(+1, -1, +1, \ldots)$

Worked: real part of $x(t) = 2e^{j\pi/3}e^{(-1+j4)t}$

1. SPLIT THE EXPONENT
$e^{(-1+j4)t} = e^{-t} \cdot e^{j4t}$

↓

2. COMBINE THE j PARTS
$e^{j\pi/3} \cdot e^{j4t} = e^{j(4t+\pi/3)}$ *(add the exponents)*

↓

3. REWRITE
$x(t) = 2e^{-t}e^{j(4t+\pi/3)}$

↓

4. EULER
$x(t) = 2e^{-t}\left[\cos(4t + \frac{\pi}{3}) + j\sin(4t + \frac{\pi}{3})\right]$

↓

5. KEEP THE REAL PART
$\mathrm{Re}\{x\} = 2e^{-t}\cos\left(4t + \frac{\pi}{3}\right)$: a damped sinusoid, envelope $2e^{-t}$

Worked: (reverse) write $5e^{-2t}\cos\left(3t - \frac{\pi}{4}\right)$ as $\mathrm{Re}\{Ce^{at}\}$

1. MATCH $|C|e^{\sigma t}\cos(\omega t + \phi)$
$|C| = 5$, $\sigma = -2$, $\omega = 3$, $\phi = -\frac{\pi}{4}$

↓

2. BUILD C
$C = |C|e^{j\phi} = 5e^{-j\pi/4}$

↓

3. BUILD a
$a = \sigma + j\omega = -2 + j3$

Worked: cosine to exponentials: $6\cos(5t)$

1. FORMULA
$\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$, $\theta = 5t$

↓

2. MULTIPLY BY 6
$6 \cdot \frac{e^{j5t}+e^{-j5t}}{2} = 3e^{j5t} + 3e^{-j5t}$

Worked: simplify $x[n] = \cos\left(\frac{9\pi}{4}n\right)$

1. REMOVE A WHOLE TURN
$\frac{9\pi}{4} - 2\pi = \frac{9\pi}{4} - \frac{8\pi}{4} = \frac{\pi}{4}$

↓

2. SAME SAMPLES
$x[n] = \cos\left(\frac{\pi}{4}n\right)$ *($2\pi n$ is a whole number of turns)*

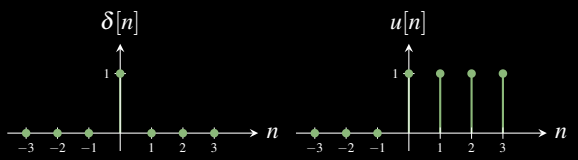
↓

3. PERIOD
$\frac{\pi/4}{2\pi} = \frac{1}{8} \Rightarrow N = 8$

10. unit impulse and unit step

DT unit impulse and **DT unit step**:

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases} \qquad u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



CT unit step $u(t)$: 0 for $t < 0$, 1 for $t > 0$. **CT unit impulse** $\delta(t)$: zero width, area 1, at $t = 0$.
Sifting: $\int_{-\infty}^{\infty} x(t)\,\delta(t-t_0)\,dt = x(t_0)$. The impulse picks out one value.
Sampling: $x(t)\,\delta(t-t_0) = x(t_0)\,\delta(t-t_0)$. DT: $x[n]\,\delta[n-n_0] = x[n_0]\,\delta[n-n_0]$.
Pulse from steps: $u(t-t_1) - u(t-t_2)$ is 1 on $[t_1, t_2)$ and 0 elsewhere.
If t_0 is outside the integration limits, the integral is 0.

Which way? — step or impulse

Given → find	Do this
$u \to \delta$ (CT)	differentiate: $\delta(t) = \dfrac{du}{dt}$
$\delta \to u$ (CT)	integrate: $u(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau$
$u \to \delta$ (DT)	first difference: $\delta[n] = u[n] - u[n-1]$
$\delta \to u$ (DT)	running sum: $u[n] = \sum_{k=-\infty}^{n} \delta[k]$

CT uses derivatives and integrals. DT uses differences and sums.

Worked: $\int_{-\infty}^{\infty} (t^2+1)\,\delta(t-2)\,dt$

1. FIND THE IMPULSE
$\delta(t-2)$ is non-zero only where $t-2=0$: $t_0 = 2$

↓

2. INSIDE THE LIMITS?
2 is inside $(-\infty, \infty)$ ✓

↓

3. EVALUATE THE REST AT t_0
$2^2 + 1 = 4 + 1 = 5$

Worked: $\int_0^5 e^t\,\delta(t+1)\,dt$

1. FIND THE IMPULSE
$t+1=0 \Rightarrow t_0 = -1$

↓

2. INSIDE THE LIMITS?
-1 is not in $[0,5] \Rightarrow$ the integral is 0

Worked: DT: $\sum_{n=-\infty}^{\infty} n^2\,\delta[n-3]$

1. FIND THE IMPULSE
$n-3=0 \Rightarrow n_0 = 3$

↓

2. EVALUATE AT n_0
$3^2 = 9$

Worked: write a pulse of height 2 on $[1,4)$ with steps, then go down and back

1. TURN ON AT $t=1$
A jump up of 2: $+2u(t-1)$

↓

2. TURN OFF AT $t=4$
A jump down of 2: $-2u(t-4)$

↓

3. ADD
$x(t) = 2u(t-1) - 2u(t-4)$

↓

4. CHECK TWO TIMES
$t=2$: $2(1) - 2(0) = 2$ ✓ $\quad t=5$: $2(1) - 2(1) = 0$ ✓

↓

5. DIFFERENTIATE (DOWN)
Each u becomes a δ: $\dfrac{dx}{dt} = 2\delta(t-1) - 2\delta(t-4)$

↓

6. INTEGRATE (BACK)
Each δ becomes a u: $2u(t-1) - 2u(t-4)$ ✓

Worked: DT: $x[n] = u[n] - u[n-3]$

1. TABLE
$n=-1$: $0-0=0$. $\quad n=0,1,2$: $1-0=1$. $\quad n=3$: $1-1=0$.

↓

2. FIRST DIFFERENCE
$u[n] - u[n-1] = \delta[n]$ and $u[n-3] - u[n-4] = \delta[n-3]$
$x[n] - x[n-1] = \delta[n] - \delta[n-3]$

Any DT signal as a sum of shifted, weighted impulses

Each stem of a DT signal is one impulse: a δ moved to the stem's position and multiplied by the stem's height.

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

- $\delta[n-k]$ is 1 only at $n=k$. So it puts a stem at position k.
- $x[k]$ is the **weight**: the height of that stem.
- At each n, every term is 0 except $k=n$. That term is $x[n]$. This is DT sifting, used the other way.

The sign inside δ is opposite to the position. A stem at $n=-1$ is $\delta[n+1]$. A stem at $n=3$ is $\delta[n-3]$.

Worked: write $x[-1]=2$, $x[0]=-1$, $x[1]=3$, $x[3]=1$ (others 0) with impulses

1. STEM AT $n=-1$, HEIGHT 2
$2\,\delta[n-(-1)] = 2\,\delta[n+1]$

↓

2. STEM AT $n=0$, HEIGHT -1
$-1 \cdot \delta[n] = -\delta[n]$

↓

3. STEM AT $n=1$, HEIGHT 3
$3\,\delta[n-1]$

↓

4. STEM AT $n=3$, HEIGHT 1
$\delta[n-3]$ *($x[2]=0$, so no term for $n=2$)*

↓

5. ADD
$x[n] = 2\delta[n+1] - \delta[n] + 3\delta[n-1] + \delta[n-3]$

Worked: (reverse) read the stems of $x[n] = 4\delta[n-2] - 2\delta[n+3]$

1. FIRST TERM
$\delta[n-2]$ is 1 at $n=2$. Stem: $x[2]=4$

↓

2. SECOND TERM
$\delta[n+3]$ is 1 at $n=-3$. Stem: $x[-3]=-2$

↓

3. CHECK AT $n=2$
$4\delta[0]-2\delta[5] = 4(1)-2(0) = 4$ ✓

Worked: a pulse as impulses: $u[n]-u[n-3]$

1. STEMS
1 at $n=0,1,2$ (section 10 table above)

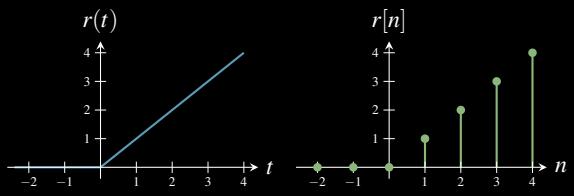
↓

2. ONE IMPULSE PER STEM
$\delta[n]+\delta[n-1]+\delta[n-2]$

The ramp signal

The **unit ramp** is 0 before time 0, then rises with slope 1:

$$r(t) = t\,u(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases} \qquad r[n] = n\,u[n] = \begin{cases} n & n \geq 0 \\ 0 & n < 0 \end{cases}$$



Other ways to write the ramp:

Form	Formula
CT: integral of the step	$r(t) = \int_{-\infty}^{t} u(\tau)\,d\tau$
CT: derivative of the ramp	$\dfrac{dr}{dt} = u(t)$
DT: running sum of the step	$r[n] = \sum_{k=-\infty}^{n-1} u[k]$
DT: first difference	$r[n] - r[n-1] = u[n-1]$
DT: sum of shifted steps	$r[n] = \sum_{k=1}^{\infty} u[n-k]$
DT: sum of impulses	$r[n] = \sum_{k=0}^{\infty} k\,\delta[n-k]$

The chain: impulse $\xrightarrow{\text{accumulate}}$ step $\xrightarrow{\text{accumulate}}$ ramp. Going back, differentiate (CT) or take the first difference (DT).

Shifted and scaled: $A\,r(t-t_0)$ is 0 before t_0, then rises with slope A.

Graphs from ramps and steps: go left to right. At each corner, add a ramp whose weight is the *change in slope*. At each jump, add a step whose weight is the *size of the jump*.

The DT sum stops at $n-1$, not n. $r[1]=1$: only $u[0]$ is added. With n as the top, you get $r[n+1]$.

Worked: check $r[n] = \sum_{k=-\infty}^{n-1} u[k]$ at $n = 0, 1, 3$

1. $n=0$
Sum of $u[k]$ for $k \leq -1$: all $0 \Rightarrow 0 = r[0]$ ✓

↓

2. $n=1$
$k \leq 0$: only $u[0]=1 \Rightarrow 1 = r[1]$ ✓

↓

3. $n=3$
$k \leq 2$: $u[0]+u[1]+u[2] = 3 = r[3]$ ✓

Worked: first difference of the ramp

1. $n=0$
$r[0]-r[-1] = 0-0 = 0$

↓

2. $n=1$
$r[1]-r[0] = 1-0 = 1$

↓

3. $n=2$
$r[2]-r[1] = 2-1 = 1$

↓

4. PATTERN
0 up to $n=0$, then 1 from $n=1$ on $\Rightarrow u[n-1]$

Worked: write a rise from 0 to 2 on $[0,2]$, then flat at 2, with ramps

1. CORNER AT $t=0$
Slope changes $0 \to 1$: change $+1 \Rightarrow +r(t)$

↓

2. CORNER AT $t=2$
Slope changes $1 \to 0$: change $-1 \Rightarrow -r(t-2)$

↓

3. ADD
$x(t) = r(t) - r(t-2)$

↓

4. CHECK $t=1$
$r(1)-r(-1) = 1-0 = 1$ ✓

↓

5. CHECK $t=5$
$r(5)-r(3) = 5-3 = 2$ ✓ *(flat at 2)*

Worked: a triangle: 0 at $t=0$, peak 1 at $t=1$, back to 0 at $t=2$

1. CORNER AT $t=0$
Slope $0 \to 1$: change $+1 \Rightarrow +r(t)$

↓

2. CORNER AT $t=1$
Slope $1 \to -1$: change $-2 \Rightarrow -2r(t-1)$

↓

3. CORNER AT $t=2$
Slope $-1 \to 0$: change $+1 \Rightarrow +r(t-2)$

↓

4. ADD
$x(t) = r(t) - 2r(t-1) + r(t-2)$

↓

5. CHECK $t=3$
$3-2(2)+1 = 0$ ✓ *(the slopes add to $1-2+1=0$)*

Worked: (reverse) sketch $x(t) = 2r(t-1) - 2r(t-3)$

1. BEFORE $t=1$
Both ramps are $0 \Rightarrow x = 0$

↓

2. FROM $t=1$ TO $t=3$
Only the first ramp: slope 2. At $t=3$: $2(3-1)=4$

↓

3. AFTER $t=3$
Slope $2-2=0$: flat at 4

↓

4. CHECK $t=5$
$2(4)-2(2) = 8-4 = 4$ ✓

Worked: a step plus a ramp: a jump of 3 at $t=0$, then slope -1 until it reaches 0

1. JUMP AT $t=0$
$+3u(t)$

↓

2. CORNER AT $t=0$
Slope $0 \to -1 \Rightarrow -r(t)$

↓

3. CORNER WHERE IT REACHES 0
$3-t=0$ at $t=3$. Slope $-1 \to 0 \Rightarrow +r(t-3)$

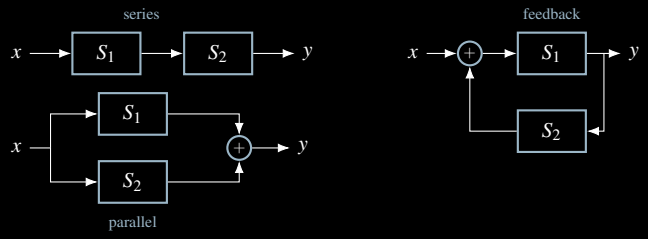
↓

4. ADD
$x(t) = 3u(t) - r(t) + r(t-3)$

11. systems and interconnections

A **system** changes an input signal into an output signal: $x(t) \to y(t)$ or $x[n] \to y[n]$.

- **Series** (cascade): the output of S_1 is the input of S_2.
- **Parallel**: x goes into both systems. The outputs add.
- **Feedback**: the output goes through S_2 and adds back to the input of S_1.



Worked: series: S_1: $w[n]=2x[n]$, then S_2: $y[n]=w[n]-w[n-1]$

1. OUTPUT OF S_1
$w[n] = 2x[n]$

↓

2. SHIFT IT FOR S_2
$w[n-1] = 2x[n-1]$ *(replace n with $n-1$)*

↓

3. SUBSTITUTE INTO S_2
$y[n] = 2x[n] - 2x[n-1]$

Worked: parallel: S_1: $2x[n]$ and S_2: $x[n-1]$

1. EACH OUTPUT
$y_1[n] = 2x[n], \quad y_2[n] = x[n-1]$

↓

2. ADD
$y[n] = 2x[n] + x[n-1]$

Worked: feedback: $y[n]=x[n]+0.5y[n-1]$, input $x[n]=\delta[n]$, $y[-1]=0$

1. $n=0$
$y[0] = \delta[0] + 0.5y[-1] = 1 + 0.5(0) = 1$

↓

2. $n=1$
$y[1] = \delta[1] + 0.5y[0] = 0 + 0.5(1) = 0.5$

↓

3. $n=2$
$y[2] = 0 + 0.5(0.5) = 0.25$

↓

4. $n=3$
$y[3] = 0 + 0.5(0.25) = 0.125$

↓

5. PATTERN
$y[n] = (0.5)^n u[n]$ *(each step multiplies by 0.5)*

DT systems as difference equations

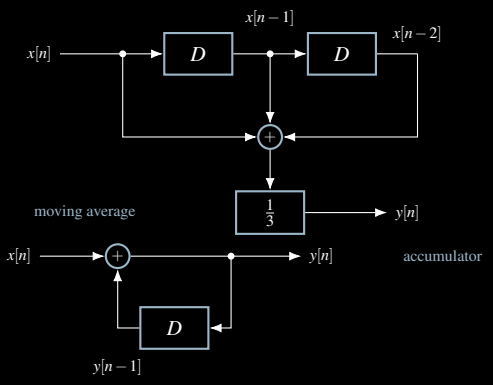
A DT system is often a **difference equation**: $y[n]$ is written with present and past inputs, and sometimes past outputs.

System	Difference equation	What it does
unit delay	$y[n] = x[n-1]$	shifts right 1
gain	$y[n] = ax[n]$	scales
first difference	$y[n] = x[n] - x[n-1]$	change since the last sample
3-point moving average	$y[n] = \frac{x[n]+x[n-1]+x[n-2]}{3}$	smooths
accumulator	$y[n] = \sum_{k=-\infty}^{n} x[k]$	running total
accumulator, recursive	$y[n] = y[n-1] + x[n]$	same running total
bank balance	$y[n] = 1.01\,y[n-1] + x[n]$	1% interest plus deposit

Nonrecursive (FIR): $y[n]$ uses inputs only. **Recursive** (IIR): $y[n]$ also uses past outputs $y[n-1], \ldots$, and needs a start value such as $y[-1]=0$.

Block diagram parts:

Part	What it does
adder (circle, $+$)	output = sum of its inputs
gain (box with a)	output = a·input
unit delay (box D)	input $x[n]$, output $x[n-1]$
branch point (dot)	copies a signal to two places



Which way? — equation or block diagram

Given → find	Do this
diagram → equation	name the signal on each wire; write the adder's output as the sum of its inputs
equation → diagram	one D per shift ($n-1$, $n-2$, …); one gain per coefficient; one adder for the sum
equation → output values	put in $n=0,1,2,\ldots$ in order, one step at a time

A delay on the output side means feedback. If a D takes $y[n]$ and feeds back into the adder, the equation has $y[n-1]$: it is recursive.

Worked: (diagram to equation) read the moving-average diagram

1. NAME THE WIRES
After the first D: $x[n-1]$. After the second D: $x[n-2]$.

↓

2. ADDER OUTPUT
$x[n] + x[n-1] + x[n-2]$

↓

3. GAIN
$y[n] = \frac{1}{3}\left(x[n] + x[n-1] + x[n-2]\right)$

Worked: (equation to diagram) $y[n] = 2x[n] - 0.5\,y[n-1]$

1. COUNT THE SHIFTS
One $y[n-1] \Rightarrow$ one D, on the output side (feedback)

↓

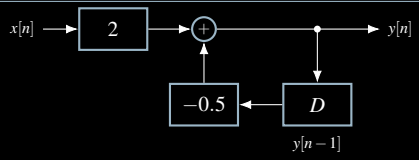
2. COUNT THE COEFFICIENTS
Gain 2 on the input. Gain -0.5 after the D.

↓

3. ONE ADDER
Its inputs: $2x[n]$ and $-0.5\,y[n-1]$. Its output: $y[n]$.

↓

4. DRAW
$x \to [2] \to (+) \to y$. From y: a dot $\to [D] \to [-0.5] \to$ back into $(+)$.



Worked: run the moving average with $x[0] = 3$, $x[1] = 6$, $x[2] = 9$, ***others*** 0

1. $n = 0$
$y[0] = \frac{1}{3}(x[0] + x[-1] + x[-2]) = \frac{1}{3}(3 + 0 + 0) = 1$

↓

2. $n = 1$
$y[1] = \frac{1}{3}(6 + 3 + 0) = \frac{9}{3} = 3$

↓

3. $n = 2$
$y[2] = \frac{1}{3}(9 + 6 + 3) = \frac{18}{3} = 6$

↓

4. $n = 3$
$y[3] = \frac{1}{3}(0 + 9 + 6) = \frac{15}{3} = 5$

↓

5. $n = 4$
$y[4] = \frac{1}{3}(0 + 0 + 9) = 3$

↓

6. $n = 5$ **AND LATER**
All three inputs are $0 \Rightarrow y = 0$

Worked: run the accumulator $y[n] = y[n-1] + x[n]$ ***with*** $x[n] = u[n]$, $y[-1] = 0$

1. $n = 0$
$y[0] = y[-1] + u[0] = 0 + 1 = 1$

↓

2. $n = 1$
$y[1] = y[0] + u[1] = 1 + 1 = 2$

↓

3. $n = 2$
$y[2] = 2 + 1 = 3$

↓

4. PATTERN
$y[n] = (n+1)\,u[n] = r[n+1]$ *(accumulating a step gives a ramp)*

Worked: (both forms agree) accumulator as a sum, $x[n] = u[n]$, ***at*** $n = 2$

1. SUM FORM
$y[2] = \sum_{k=-\infty}^{2} u[k] = u[0] + u[1] + u[2]$

↓

2. ADD
$1 + 1 + 1 = 3$, the same as the recursive form ✓

12. system properties

- **Memoryless**: y at time n uses only x at time n.
- **Causal**: y at time n uses no x from a later time.
- **Stable** (BIBO): each bounded input gives a bounded output.
- **Invertible**: different inputs give different outputs. You can solve for x.
- **Time invariant**: a delay in x gives the same delay in y.
- **Linear**: superposition holds. If $x_1 \to y_1$ and $x_2 \to y_2$, then $ax_1 + bx_2 \to ay_1 + by_2$.

Quick linearity test: a linear system gives $y = 0$ for $x = 0$. If it does not, it is not linear.
Hidden time variance: a t or n outside of $x(\cdot)$, or inside it as $x(2t)$ or $x(-t)$, is a warning sign.
$y(t) = x(-t)$ is not causal: at $t = -1$ it needs $x(1)$, a later time.

Worked: memory and causality of $y[n] = x[n] + x[n-1]$ ***and*** $y[n] = x[n+1]$

1. $y[n] = x[n] + x[n-1]$
It uses $x[n-1]$, a past value. has memory, causal

↓

2. $y[n] = x[n+1]$
At $n = 0$ it needs $x[1]$, a later value. not causal

Worked: stability of $y[n] = \sum_{k=-\infty}^{n} x[k]$

1. PICK A BOUNDED INPUT
$x[n] = u[n]$. It is never larger than 1.

↓

2. COMPUTE THE OUTPUT
$y[n] = \sum_{k=0}^{n} 1 = n + 1$ for $n \geq 0$

↓

3. IS IT BOUNDED?
$n + 1$ grows without limit $\Rightarrow$ not stable

Worked: invertibility of $y = 2x$ ***and*** $y = x^2$

1. $y = 2x$
Solve: $x = \dfrac{y}{2} \Rightarrow$ invertible

↓

2. $y = x^2$
$x = 1$ and $x = -1$ both give $y = 1 \Rightarrow$ not invertible

Worked: time invariance of $y[n] = n\,x[n]$

1. DELAY THE INPUT, THEN APPLY THE SYSTEM
New input $x_2[n] = x[n - n_0]$. Output: $y_2[n] = n\,x[n - n_0]$

↓

2. APPLY THE SYSTEM, THEN DELAY THE OUTPUT
Replace each n in $y[n] = n\,x[n]$ with $n - n_0$: $y[n - n_0] = (n - n_0)\,x[n - n_0]$

↓

3. COMPARE
$n\,x[n - n_0] \neq (n - n_0)\,x[n - n_0] \Rightarrow$ time varying

Worked: time invariance of $y(t) = x(2t)$

1. DELAY THE INPUT, THEN APPLY THE SYSTEM
$x_2(t) = x(t - t_0)$. $y_2(t) = x_2(2t) = x(2t - t_0)$ *(put $2t$ where t was in x_2)*

↓

2. APPLY THE SYSTEM, THEN DELAY THE OUTPUT
$y(t - t_0) = x\big(2(t - t_0)\big) = x(2t - 2t_0)$

↓

3. COMPARE
$x(2t - t_0) \neq x(2t - 2t_0) \Rightarrow$ time varying

Worked: time invariance of $y[n] = x[n-1]$

1. DELAY THE INPUT, THEN APPLY THE SYSTEM
$y_2[n] = x_2[n-1] = x[n - 1 - n_0]$

↓

2. APPLY THE SYSTEM, THEN DELAY THE OUTPUT
$y[n - n_0] = x[n - n_0 - 1]$

↓

3. COMPARE
The same $\Rightarrow$ time invariant

Worked: linearity of $y[n] = 2x[n] + 3$

1. QUICK TEST
$x = 0$ gives $y = 3 \neq 0 \Rightarrow$ not linear. The full test confirms it:

↓

2. OUTPUT FOR THE COMBINED INPUT
$x_3 = ax_1 + bx_2$. $y_3 = 2(ax_1 + bx_2) + 3 = 2ax_1 + 2bx_2 + 3$

↓

3. COMBINATION OF THE OUTPUTS
$ay_1 + by_2 = a(2x_1 + 3) + b(2x_2 + 3) = 2ax_1 + 2bx_2 + 3a + 3b$

↓

4. COMPARE
$3 \neq 3a + 3b$ for most $a, b \Rightarrow$ not linear *(linear plus a constant: incrementally linear)*

Worked: linearity of $y[n] = nx[n]$

1. OUTPUT FOR THE COMBINED INPUT
$y_3 = n\big(ax_1[n] + bx_2[n]\big) = anx_1[n] + bnx_2[n]$

↓

2. COMBINATION OF THE OUTPUTS
$ay_1 + by_2 = anx_1[n] + bnx_2[n]$

↓

3. COMPARE
The same $\Rightarrow$ linear *(linear, but time varying: the properties are separate)*

Stability with a bound B — the procedure

BIBO stable: each bounded input gives a bounded output. Bounded means there is a finite number B with

$$|x[n]| \leq B < \infty \quad \text{for every } n$$

To show stable:

1. Assume $|x[n]| \leq B$ for every n. Then each shifted input is also $\leq B$: $|x[n-1]| \leq B$.
2. Take the absolute value of the equation.
3. Use $|a+b| \leq |a| + |b|$ and $|cx| = |c|\,|x|$.
4. Replace each $|x[\cdot]|$ with B.
5. If the result is a finite number (no n left, no infinite sum), the system is stable.

To show unstable: find one bounded input (for example $x[n] = 1$ or $u[n]$) whose output grows without limit.

An n that multiplies x means the bound fails. $|nx[n]| \leq |n|B$ grows with n. A sum with no end can fail. The accumulator adds B again at each step.

Worked: is $y[n] = 2x[n] + x[n-1]$ stable?

1. ASSUME A BOUND
$|x[n]| \leq B$ for every n, so $|x[n-1]| \leq B$ too

↓

2. ABSOLUTE VALUE
$|y[n]| = |2x[n] + x[n-1]|$

↓

3. SPLIT THE SUM
$\leq 2|x[n]| + |x[n-1]|$

↓

4. PUT IN B
$\leq 2B + B = 3B$

↓

5. CONCLUDE
$3B$ is finite $\Rightarrow$ stable

Worked: is the 3-point moving average stable?

1. ABSOLUTE VALUE AND SPLIT
$|y[n]| \leq \frac{1}{3}\big(|x[n]| + |x[n-1]| + |x[n-2]|\big)$

↓

2. PUT IN B
$\leq \frac{1}{3}(B + B + B) = B \Rightarrow$ stable

Worked: is $y[n] = nx[n]$ stable?

1. TRY A BOUNDED INPUT
$x[n] = 1$ for all n. $B = 1$

↓

2. OUTPUT
$y[n] = n \cdot 1 = n$

↓

3. CONCLUDE
n grows without limit $\Rightarrow$ not stable *(one counterexample is enough)*

Worked: is $y[n] = e^{x[n]}$ stable?

1. ASSUME A BOUND
$-B \leq x[n] \leq B$

↓

2. $e^{(\cdot)}$ INCREASES
$e^{-B} \leq e^{x[n]} \leq e^{B}$

↓

3. CONCLUDE
$|y| \leq e^{B}$, finite $\Rightarrow$ stable

Worked: (recursive) is $y[n] = 0.5\,y[n-1] + x[n]$ stable, with $y[-1] = 0$?

1. UNROLL THE RECURSION
$y[0] = x[0]$. $y[1] = x[1] + 0.5x[0]$. $y[2] = x[2] + 0.5x[1] + 0.25x[0]$.

↓

2. PATTERN
$y[n] = \sum_{k=0}^{n} (0.5)^k x[n-k]$

↓

3. ABSOLUTE VALUE AND B
$|y[n]| \leq \sum_{k=0}^{n} (0.5)^k B \leq B \sum_{k=0}^{\infty} (0.5)^k$

↓

4. GEOMETRIC SUM
$$\sum_{k=0}^{\infty} (0.5)^k = \frac{1}{1 - 0.5} = 2$$

↓

5. CONCLUDE
$|y| \leq 2B \Rightarrow$ stable *(with 1.01 in place of 0.5, the sum has no limit: unstable)*

Properties of common DT systems

System	Mless	Causal	Linear	TI	Stable
$x[n-1]$	no	yes	yes	yes	yes
$x[n] - x[n-1]$	no	yes	yes	yes	yes
moving average	no	yes	yes	yes	yes
accumulator	no	yes	yes	yes	no
$nx[n]$	yes	yes	yes	no	no
$x[n]^2$	yes	yes	no	yes	yes
$2x[n] + 3$	yes	yes	no	yes	yes
$x[n+1]$	no	no	yes	yes	yes
$x[2n]$	no	no	yes	no	yes
$x[-n]$	no	no	yes	no	yes
$0.5\,y[n-1] + x[n]$	no	yes	yes	yes	yes

Mless = memoryless. TI = time invariant. The recursive system starts at rest ($y[-1] = 0$).

The properties are separate. A system can be linear but not time invariant ($nx[n]$), or stable but not causal ($x[n+1]$).